# Ji 2020 · Cutaneous SCC — cell-state recapitulation

## GO / NO-GO GATES

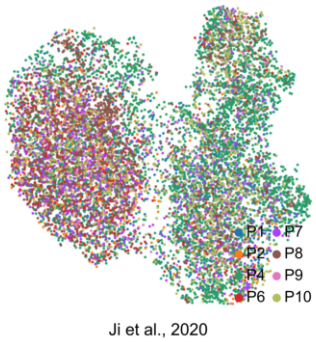
**Gate 1** **GO** Malignant cells from 8/10 samples (≥100 cells) form one continuous, sample-mixed manifold (iLISI≈4.0; single kNN component; 13 detached cells removed). Cell-cycle regression was required — it is the largest program and otherwise dominates the embedding.

**Gate 2** **GO** The embryonic modules resolve into a branching wishbone: a Gast/pSom co-expression hub diverges into a Gast-leaning (Basal) arm and a pSom-leaning (Differentiated) arm. Crest is a terminal readout at the tips, uncoupled from Gast/pSom.

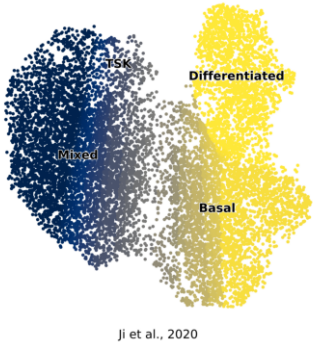
**Gate 3** **GO** Within the 4,888-cell Mixed hub, per-gene Gast and pSom loadings mutually anti-correlate: r = −0.54 (Puram anchor ≈ −0.49) — competitive cross-repression, the bifurcation signature.

## STEP FIGURES

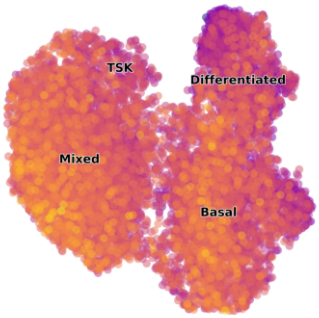
01 · Sample integration (Gate 1)



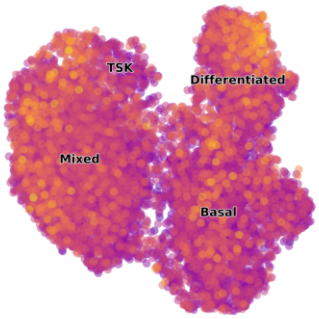

02 · Trajectory + root (hub)



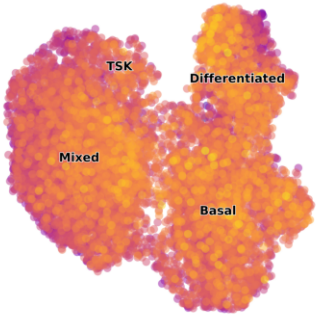

03 · Gast signature



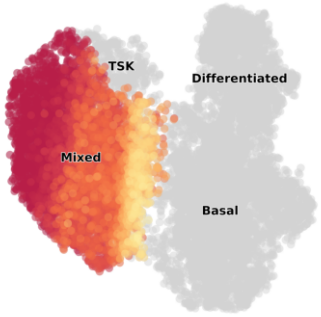

03 · pSom signature



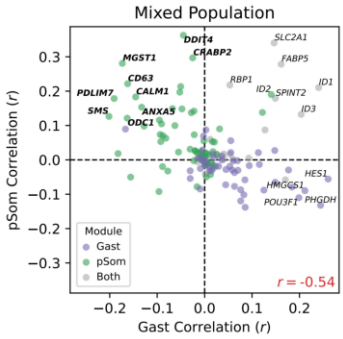

03 · Crest signature



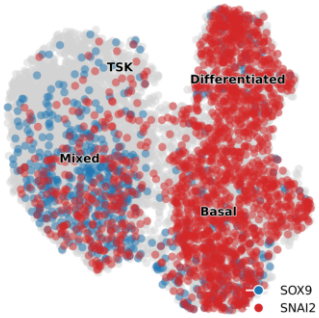

04 · Hub module correlation




05 · Cross-repression r=−0.54




06 · EMT-TFs (SOX9 / SNAI2)




**Dataset.** Cutaneous SCC (Ji et al., 2020) · 10x · 10 patients, 48,164 cells; 10,598 malignant → 10,518 after QC · subtypes Mixed / Basal / Differentiated / TSK.

**Verdict.** ✓ Pattern reproduces — all three gates GO. Branching wishbone rooted at the Gast/pSom hub; competitive Gast↔pSom cross-repression (r=−0.54); Crest terminal, uncoupled.

## PARAMETER CHOICES

| | |
|---|---|
| **Min cells / sample** | ≥ 100 (gap 40→268, 6.7×; keeps 8/10, drops P3, P5) |
| **Cell-cycle regression** | S + G2M scores regressed — KEY to a clean manifold |
| **Island removal** | UMAP radius-graph, largest component (−13 cells) |
| **Integration** | Harmony on sample, θ = 2, 20 iterations; n_pcs = 20 |
| **Embedding** | UMAP min_dist = 0.4, spread = 1.0 |
| **Trajectory** | scFates PPT tree, Nodes = 60, λ = 2000, σ = 1.5 |
| **Geometry / root** | Branching (3-tip wishbone); root = Mixed hub (node 16, a tip) |
| **Hub / Gate-3 r** | Mixed (4,888 cells); r = −0.54 |
| **EMT-TFs shown** | SOX9 (hub/basal), SNAI2 (Differentiated arm) — Snail/Zeb/Twist ~absent |
| **Subtype annotation** | Leiden markers + Ji 2020 KC hierarchy (Basal/TSK/Diff) |